The freezing point of the solution is much lower than the solvent as the temperature at which the solid and the liquid have the same vapour pressure is much lower than the solvent.

- (d) Osmotic pressure: The spontaneous flow of solvent molecules from a dilute solution into a concentrated solution when the two are separated by a perfect semipermeable membrane is called osmosis.
- Osmotic pressure (π) is the pressure which must be applied to the solution side (more concentrated solution) to just prevent the passage of pure solvent into it through a semipermeable membrane.
- Abnormal colligative properties
- If solutes undergo any association or dissociation in a solutiont, they exhibit abnormal colligative properties Van't Hoft factor explains quantitatively the extent of association or dissociation of soluts in solvent.

Evaluation

I. Choose the best answer.

1. The molality of a solution containing 1.8g of glucose dissolved in 250g of water is

   a) 0.2 M b) 0.01 M

   c) 0.02 M d) 0.04 M

2. Which of the following concentration terms is / are independent of temperature

   a) molality b) molarity

   c) mole fraction d) (a) and (c)

3. Stomach acid, a dilute solution of HCl can be neutralised by reaction with Aluminium hydroxide

   $Al(OH)_3 + 3HCl\ (aq) \rightarrow AlCl_3 + 3\ H_2O$

   How many millilitres of 0.1 M $Al(OH)_3$ solution are needed to neutralise 21 mL of 0.1 M HCl ?

   a) 14 mL b) 7 mL

   c) 21 mL d) none of these

4. The partial pressure of nitrogen in air is 0.76 atm and its Henry's law constant is 7.6×10^4 atm at 300K. What is the molefraction of nitrogen gas in the solution obtained when air is bubbled through water at 300K ?

   a) 1×10^{-4} b) 1×10^{-6}

   c) 2×10^{-5} d) 1×10^{-5}

5. The Henry's law constant for the solubility of Nitrogen gas in water at 350 K is 8×10^4 atm. The mole fraction of nitrogen in air is 0.5. The number of moles of Nitrogen from air dissolved in 10 moles of water at 350K and 4 atm pressure is

   a) 4×10^{-4} b) 4×10^4

   c) 2×10^{-2} d) 2.5×10^{-4}

6. Which one of the following is incorrect for ideal solution ?

   a) $\Delta H_{mix} = 0$

   b) $\Delta U_{mix} = 0$

   c) $\Delta P = P_{observed} - P_{Calculated\ by\ raoults\ law} = 0$

   d) $\Delta G_{mix} = 0$

7. Which one of the following gases has the lowest value of Henry's law constant ?

   a) N_2 b) He

   c) CO_2 d) H_2

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